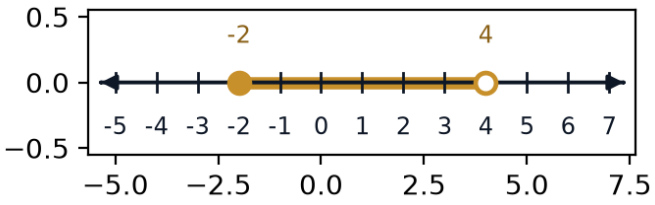
Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving contextual problems with integers

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(5)$.

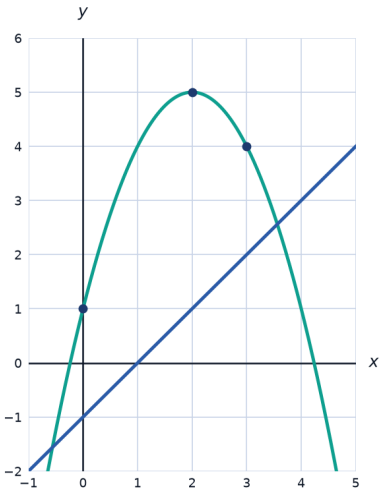


2 [Foundational] • Target: Solving contextual problems with integers

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

3 [Foundational] • Target: Fractions and the number line

For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(2)$.



4 [Foundational] • Target: Fractions and the number line

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

5 [Foundational] • Target: Fractions and the number line

A data set recorded for fractions and the number line consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

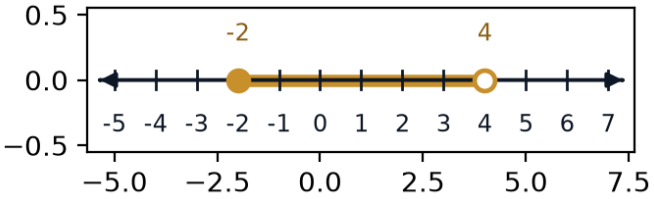
6 [Foundational] • Target: Fractions and the number line

For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(3)$.

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7 [Foundational] • Target: Fractions and the number line

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Fractions and the number line

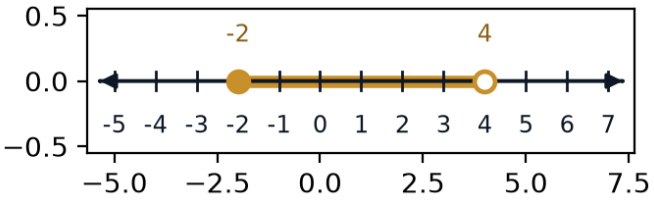
For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(4)$.

9 [Intermediate] • Target: Fractions and the number line

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

10 [Intermediate] • Target: Fractions and the number line

For the model $f(x) = 5x + 6$ in fractions and the number line, evaluate $f(5)$.

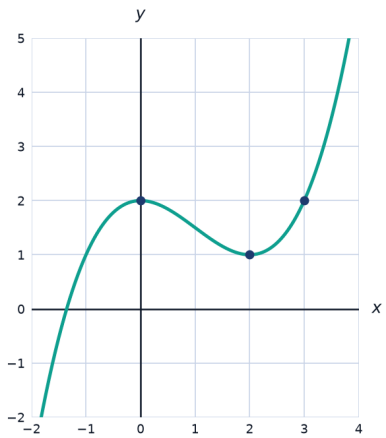


11 [Intermediate] • Target: Fractions and the number line

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

12 [Intermediate] • Target: Multiplying and dividing integers

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(2)$.



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13 [Intermediate] • Target: Multiplying and dividing integers

A data set recorded for multiplying and dividing integers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Multiplying and dividing integers

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(3)$.

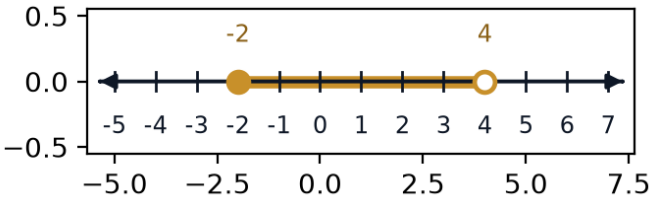
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Multiplying and dividing integers • Cross-Lesson: 10.01 Sets & Venn Diagrams

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(4)$.

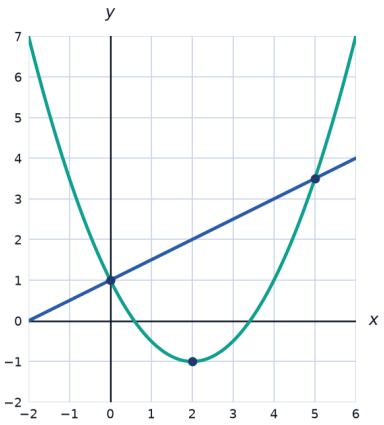
16 [Challenging] • Target: Multiplying and dividing integers • Cross-Lesson: 2.01 Algebra & Graphs

For the model $f(x) = 6x + 7$ in multiplying and dividing integers, evaluate $f(5)$.



17 [Challenging] • Target: Mixed operations with integers • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(2)$.



18 [Challenging] • Target: Mixed operations with integers • Cross-Lesson: 4.01 Mensuration

A data set recorded for mixed operations with integers consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Mixed operations with integers • Cross-Lesson: 6.01 Trigonometry

For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(3)$.

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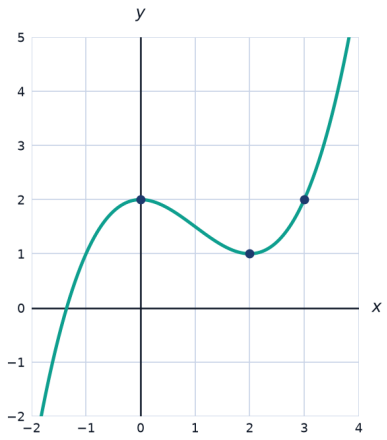
20 [Challenging] • Target: Mixed operations with integers • Cross-Lesson: 7.01 Transformations & Vectors

For the model $f(x) = 2x + 8$ in mixed operations with integers, evaluate $f(4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Integers and the number line

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(2)$.



a. 7

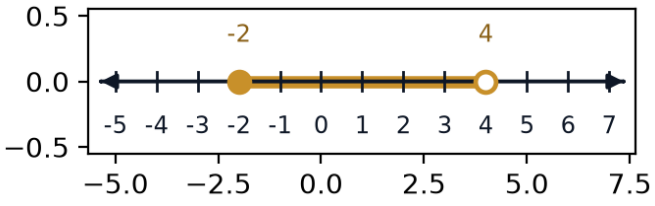
b. 9

c. 4

d. 11

22 [Foundational] • Target: Integers and the number line

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

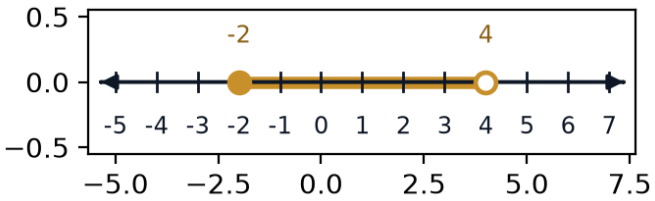
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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23 [Foundational] • Target: Integers and the number line

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(5)$.



a. 13

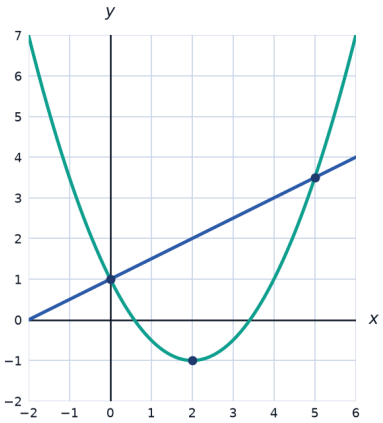
b. 15

c. 10

d. 17

24 [Foundational] • Target: Adding and subtracting integers

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(2)$.



a. 10

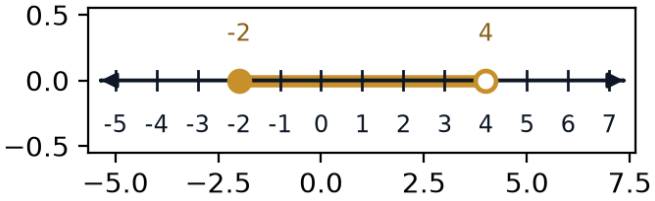
b. 13

c. 6

d. 14

25 [Foundational] • Target: Adding and subtracting integers

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

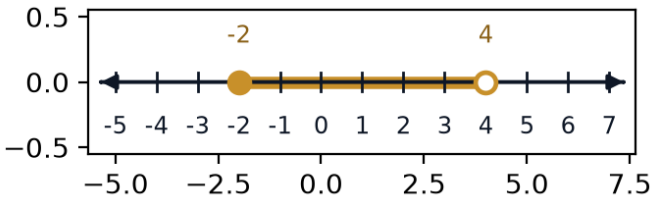
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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26 [Foundational] • Target: Adding and subtracting integers

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(5)$.



a. 19

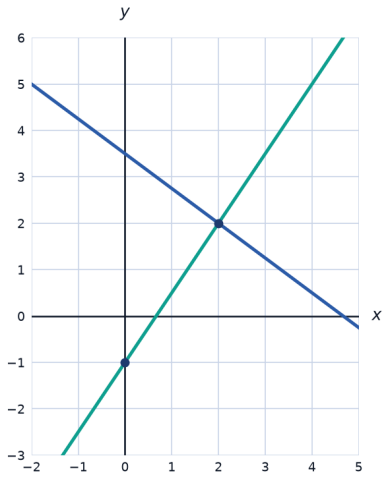
b. 22

c. 15

d. 23

27 [Foundational] • Target: Solving contextual problems with integers

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(2)$.



a. 13

b. 17

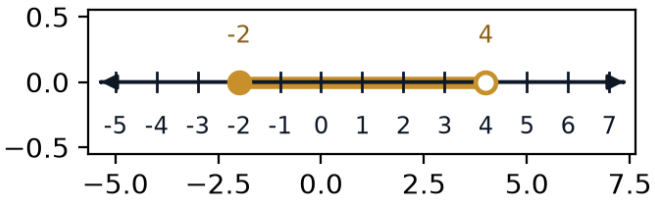
c. 8

d. 17

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28 [Foundational] • Target: Solving contextual problems with integers

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Integers and the number line

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

30 [Intermediate] • Target: Integers and the number line

A data set recorded for integers and the number line consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

31 [Intermediate] • Target: Integers and the number line

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

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32 [Intermediate] • Target: Integers and the number line

For the model $f(x) = 2x + 3$ in integers and the number line, evaluate $f(4)$.

a. 11

b. 13

c. 8

d. 15

33 [Intermediate] • Target: Integers and the number line

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

34 [Intermediate] • Target: Integers and the number line

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

35 [Intermediate] • Target: Adding and subtracting integers

A data set recorded for adding and subtracting integers consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

36 [Intermediate] • Target: Adding and subtracting integers

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

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37 [Intermediate] • Target: Adding and subtracting integers

For the model $f(x) = 3x + 4$ in adding and subtracting integers, evaluate $f(4)$.

a. 16

b. 19

c. 12

d. 20

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Adding and subtracting integers • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

39 [Challenging] • Target: Adding and subtracting integers • Cross-Lesson: 6.01 Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

40 [Challenging] • Target: Solving contextual problems with integers • Cross-Lesson: 7.01 Transformations & Vectors

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

41 [Challenging] • Target: Solving contextual problems with integers • Cross-Lesson: 8.01 Probability & Statistics

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

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42 [Challenging] • Target: Solving contextual problems with integers • Cross-Lesson: 9.01 Functions

For the model $f(x) = 4x + 5$ in solving contextual problems with integers, evaluate $f(4)$.

a. 21

b. 25

c. 16

d. 25

43 [Challenging] • Target: Solving contextual problems with integers • Cross-Lesson: 1.01 Number

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

44 [Challenging] • Target: Fractions and the number line • Cross-Lesson: 10.01 Sets & Venn Diagrams

A data set recorded for fractions and the number line consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

45 [Challenging] • Target: Multiplying and dividing integers • Cross-Lesson: 2.01 Algebra & Graphs

A data set recorded for multiplying and dividing integers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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